Form No. EE/CP/A-01

Daffodil International University
Department of Electrical and Electronic Engineering (EEE)

OPEN ENDED EXPERIMENT

Title or Problem Statement: Design of a DC Resistive Network and an RC Low-Pass Filter		
Course Title: Electrical Circuits Laboratory		**Course Code: 0713-122**
Student Name: Tanvir Ahmmed		**Student ID: 261-33-005**
Instructor's Name: MD Wahiduzzaman		**Submission Date: 14-08-2026**
Batch: 45	**Semester: Summer 2026**	**Level-Term: 1-2**
Program: B.Sc. in Electrical and Electronic Engineering		

Signature of the Instructor

Assessment Mapping (Instructor Use Only)

Task Statement	CO No.	Mapped PO	K	WP	Domain/Level C/A/P	Assigned Marks	Obtained Marks
Design and build a DC circuit using two voltage sources and at least three resistors sharing one common load resistor, and an RC low-pass filter using an available resistor and capacitor on the breadboard.	CO1	PO5.c	K6	P1 P3	C3, P3	15	
Working as a team, take measurements with the multimeter, ammeter, voltmeter, function generator and oscilloscope; verify KVL, KCL and the Superposition Theorem for the DC network, find the critical (cut-off) frequency of the RC filter, and compare all results with theory.	CO2	PO9.b	-	P1 P3	C4, A4	10	

Course Code:	0	7	1	3	-	1	2	2	Course Title:	Electrical Circuits Laboratory

Addressing the Range of Complex Engineering Problem Solving (WP), For P1-7

Attribute (WP)	Mapped	Please write your comments against each attribute mapped in the space below. Attach more sheets if necessary. (Optional)
P1: Depth of knowledge required	√	A strong understanding of fundamental circuit theory is required to attempt this problem. For the DC network, students must apply Ohm's law, Kirchhoff's voltage and current laws, series-parallel resistance reduction and the superposition principle for a linear circuit with two independent sources. For the RC filter, they must

Attribute (WP)	**Mapped**	**Please write your comments against each attribute mapped in the space below. Attach more sheets if necessary. (Optional)**
		understand capacitive reactance, the behaviour of a first-order low-pass filter and the meaning of the critical (cut-off) frequency. Correct operation of the DC supply, multimeter, ammeter, voltmeter, function generator and oscilloscope is also essential.
P2: Range of conflicting requirements		
P3: Depth of analysis required	√	In-depth analysis is needed to select suitable component values from the available kit, predict the branch currents and node voltages of the two-source network, and separate the load current into the contributions of each source to apply superposition. For the filter, students must calculate the theoretical cut-off frequency from their chosen R and C, then locate it experimentally from the frequency response (where the output falls to about 70% of the input). Comparing theoretical and experimental results and explaining discrepancies caused by component tolerance and meter loading requires substantial analytical reasoning.
P4: Familiarity of issues		
P5: Extent of applicable codes		
P6: Extent of stakeholder involvement and conflicting requirements		
P7: Interdependence		

Students are required to complete the **P1** and **P3** justification sections. Sample justifications have been provided as examples to help you understand the expected format, level of detail, and writing style. Use these examples as a guide when preparing your own justifications.

Assessment Rubric

Performance Indicator	Excellent (100%-85%)	Very Good (84%-70%)	Good (69%-55%)	Fair (54%-40%)	Poor (39%-1%)
Technical Accuracy & Analysis	Accurate values (R, C, cut-off freq., currents), correct KVL/KCL/superposition and filter calculations with proper units; correct methodology clearly followed	Minor calculation errors but correct methodology and interpretation	Partial correctness; weak interpretation	Incorrect method or missing key steps	Completely incorrect analysis
Experimental Design & Implementation	Proper design/setup, correct instruments, safe implementation	Minor issues in setup but overall correct	Partially correct setup	Poor design; major components missing	No proper design
Data Analysis & Interpretation	Clear data, graphs, strong interpretation	Good data with minor issues	Basic analysis but lacks depth	Weak or incorrect interpretation	No meaningful analysis
Organization & Presentation	Well-organized, clear sections, neat figures	Generally organized with minor issues	Somewhat organized	Poor organization	Disorganized
Timeliness	On time	Late with excuse	Very late with excuse	Missed deadline, needed reminder	Not submitted
Completion of Work	All tasks completed	One part missing	More than one part missing	Multiple major parts missing	Mostly incomplete

Knowledge Profile

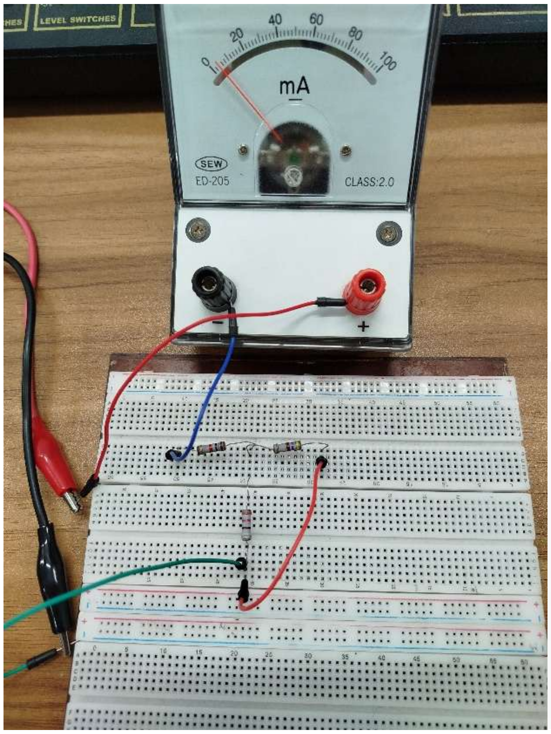
Code	Attribute	Easy Keywords
K1	Natural Science Knowledge	Physics, Chemistry, Basic Science
K2	Mathematics and Computing	Math, Statistics, Programming, Logic
K3	Engineering Fundamentals	Ohm's Law, Kirchhoff's Rules, Basic Circuits
K4	Specialized Engineering Knowledge	Advanced Electronics, Control Systems, Power Systems
K5	Engineering Design	Circuit Design, System Design, Problem Solving

Code	Attribute	Easy Keywords
K6	Engineering Tools and Practice	Lab Work, Simulation Tools (e.g. Pspice, MATLAB), Hands-on
K7	Engineering and Society	Safety, Environment, Ethics, Social Impact
K8	Research Literature	Reading Journals, Research Papers, Final Year Projects

Three Learning Domain

Cognitive Domain (PO1-PO8, PO11), Head (Written)	Affective Domain (PO6-PO12), Heart (Feelings)	Psychomotor Domain (PO4-PO5), Hand (Practical)
C1: Remembering	A1: Receive	P1: Imitate
C2: Understanding	A2: Respond	P2: Manipulation
C3: Applying	A3: Value	P3: Precision
C4: Analyzing	A4: Organize	P4: Articulation
C5: Evaluating	A5: Internalize	P5: Naturalization
C6: Creating/Designing		

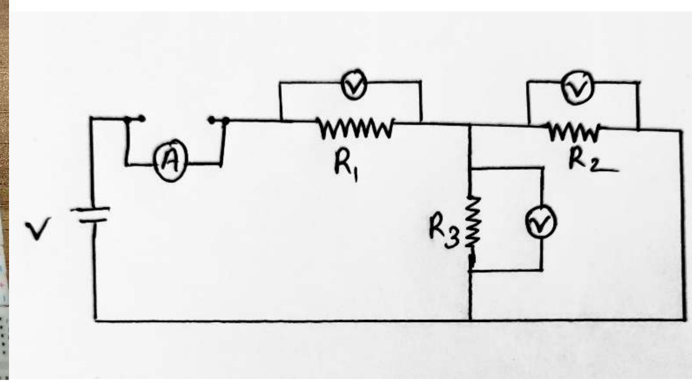
Kirchhoff's Voltage Law (KVL): KVL states that around any closed circuit the sum of the voltage rises equals the sum of the voltage drops across the resistors.



Σ VRISES = Σ VDROPS

VS = V1 + V2 + V3 +. . . . .

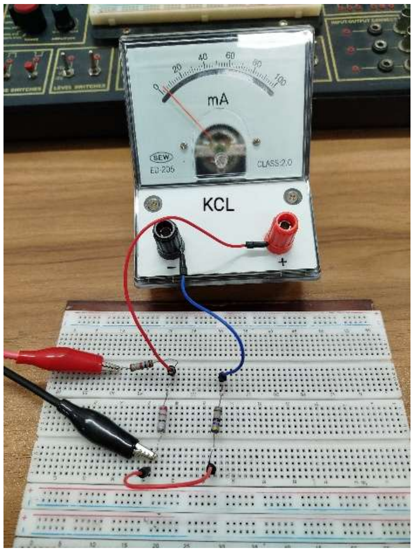
R_1= 1000 Ω
R_2= 464 Ω
R_3= 216 Ω

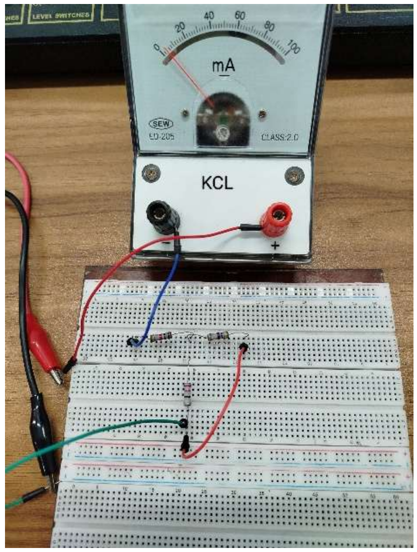


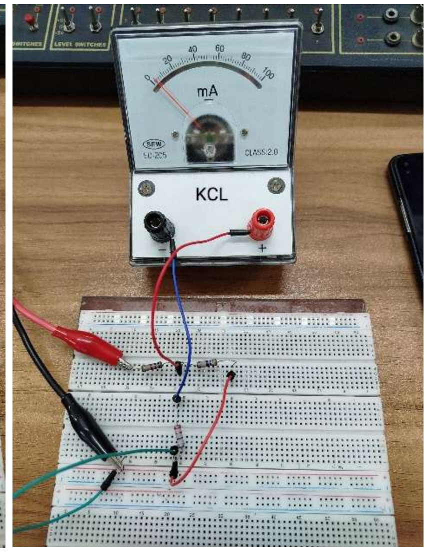
Data for KVL

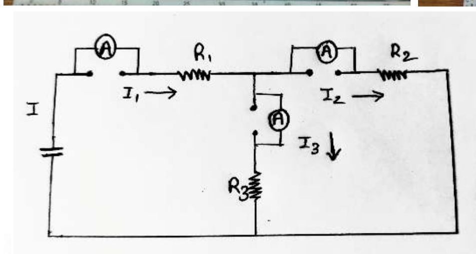
	DC supply Display		Multimeter				Calculation			
obs. No.	V_s (V)	I_s (mA)	V_1 (V)	V_2 (V)	V_3 (V)	$R_s = R_1+(R_2 \parallel R_3)$ (Ω) Experimental	$R_{SE} = \frac{V_s}{I_s}$ (Ω) Experimental	$V_{SE} = V_1 + V_2$ (V) Experimental	Error $V_s - V_{SE}$	Error $R_s - R_{SE}$
1	2.8	2.4	2.40	0.39	0.39	1197.79	1166.67	2.79	0.357%	1.695%
2	3.6	3.1	3.10	0.49	0.49	1197.79	1161.29	3.59	0.278%	1.176%
3	5.1	4.4	4.40	0.69	0.69	1197.79	1159.09	5.09	0.196%	0.985%
4	7.3	6.4	6.40	0.88	0.88	1197.79	1140.63	7.28	0.274%	0.624%
5	9.8	8.5	8.50	1.28	1.28	1197.79	1152.94	9.78	0.204%	0.499%

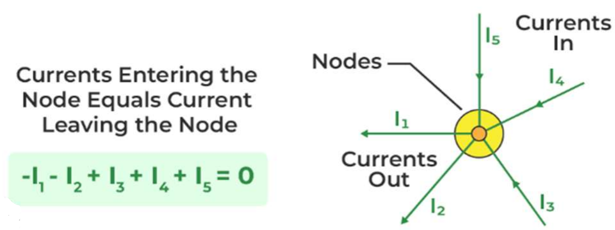
Verification of Kirchhoff's Current Law (KCL): KCL states that the total current entering a node or junction in an electrical circuit must equal the total current leaving that same node.







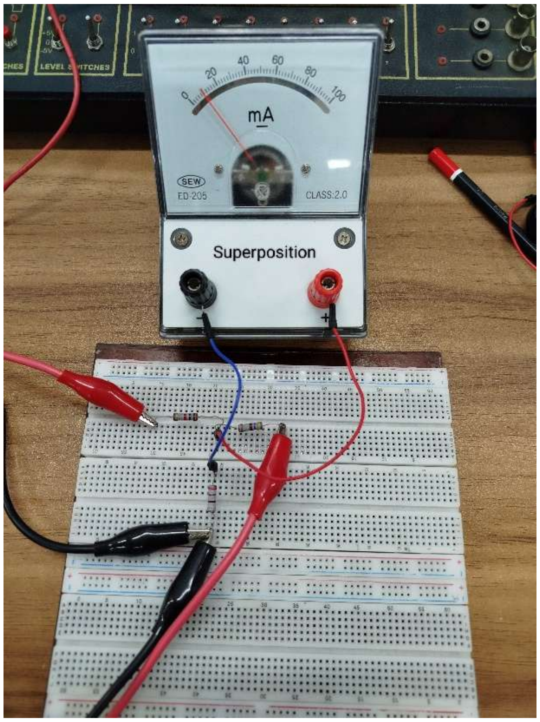


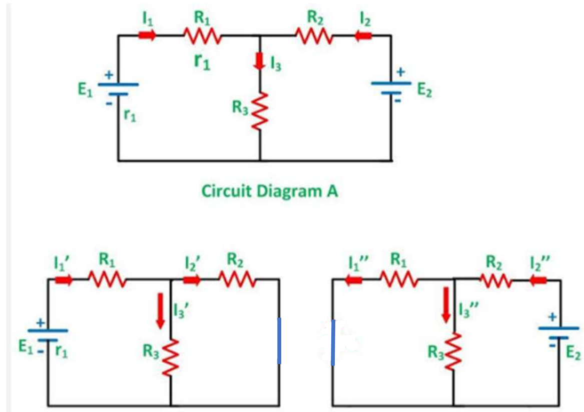


KCL

Obs No.	DC Supply		Ammeter (mA)			$I_E = I_2 + I_3$ (A) Experimental	R_P (Ω) Theoritical	$R_{PE} = V_s/I_s$ (Ω) Experimental	Error $I - I_E$	Error $R_P - R_{PE}$
	V_s (V)	I_s (mA)	I_1	I_2	I_3					
1	4	3.5mA	3.5	1.1	2.3	3.4	1147.79	1142.85	2.85%	0.43%
2	6	5.2mA	5.2	1.5	3.5	5	1147.79	1153.84	3.84%	0.59%
3	8	7mA	7	2.2	4.7	6.9	1147.79	1142.85	1.43%	0.36%
4	10	8.6mA	8.6	2.6	5.9	8.5	1147.79	1162.79	1.16%	1.37%
5	12	10mA	10	3.3	7.1	10.4	1147.79	1200	4%	4.62%

Superposition Theorem: The superposition theorem states that in any linear circuit with multiple independent power sources, the total voltage or current at any given point is equal to the algebraic sum of the individual voltages or currents produced by each independent source acting alone.





Circuit Diagram A

Circuit Diagram B

Circuit Diagram C

R_1= 1000 Ω

R_2= 464 Ω

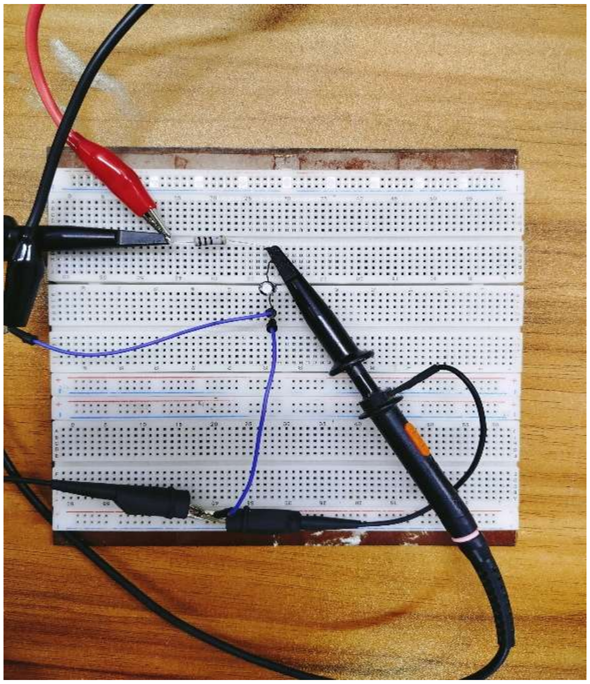
R_3= 216 Ω

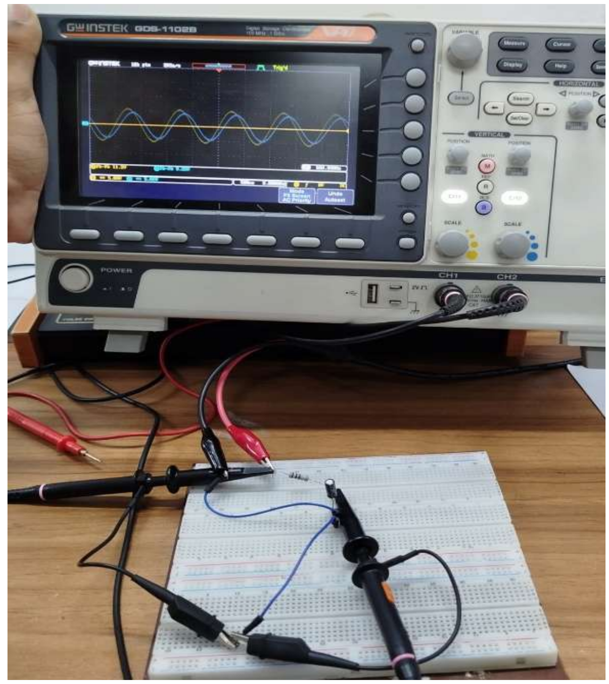
Data for superposition

No. of obj.	E_1	E_2	I (with Both E_1 and E_2 active) (A)	I′ (with only E_1 active) (A)	I″ with only E_2 active (A)	$I_{Exp} = I' + I''$ (A) Experimental	Error $I - I_{Exp}$
1	5	5	8.4×10^{-3}	5.0×10^{-3}	3.1×10^{-3}	8.1×10^{-3}	0.3
2	6	7	9.9×10^{-3}	6.3×10^{-3}	4.4×10^{-3}	10.7×10^{-3}	0.8
3	8	9	12.3×10^{-3}	8.5×10^{-3}	4×10^{-3}	12.5×10^{-3}	0.2

RC Low-Pass Filter:

An RC low-pass filter is a simple electronic circuit made of a resistor (R) and a capacitor (C). It lets low-frequency signals pass through to the output while blocking or reducing high-frequency.





Data for RC Low pass filter

No. of Obs.	Frequency(f) (100Hz - 10KHz)	Input voltage (V_i) Constant	Output voltage (V_o)	$A_o = \frac{V_o}{V_i}$
1	100	10.2	10.2	1
2	200	10.2	10	0.98
3	500	10.2	9.6	0.94
4	1000	10.2	8.80	0.86
5	1590	10.2	7.20	0.70
6	2000	10.2	5.5	0.54
7	3000	10.2	4.6	0.45
8	4000	10.2	3.6	0.35
9	6000	10.2	2.9	3.52
10	8000	10.2	2.2	0.22

Filter: Low pass Filter (LPF)

$R = 100\,\Omega$

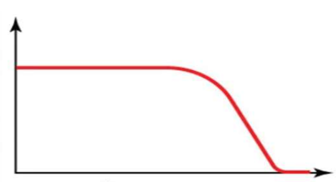
$C = 1\,\mu F$

$\therefore$ Cut-off frequency, $f_c = \frac{1}{2\pi RC}$

$= \frac{1}{2\pi \times 100 \times (1 \times 10^{-6})}$

$= 1591.55$ Hz.

Experimental, $f_c = 1590$ Hz.



Low-pass